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Research paper

Bi-level optimization of novel distribution network with VPP and flexible load cluster

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ABSTRACT

To satisfy the power requirements of modern industrial parks, while also enhancing energy efficiency and system economy, a bi-level optimization model is proposed for a novel distribution network, which incorporates a virtual power plant (VPP) and flexible load cluster. The proposed optimization model for the distribution network layer considers factors such as minimum voltage offset and operating cost. It accounts for uncertainties associated with new energy sources, control of the soft open point (SOP), as well as dispatching of the static var compensator (SVC). A normalized normal constraint (NNC) method is used to obtain Pareto front. The power flow model is converted into a normalized cone programming model utilizing second-order cone relaxation technique. The VPP optimization layer aims to achieve the lowest operational cost and comprehensively considers the complementarity of multiple types of distributed energy resources. The bi-level iterative solution is obtained using the analytical target cascading (ATC) method. Finally, the enhanced IEEE 33-node system example demonstrates that the proposed bi-level optimization model effectively reduces the system's operating cost, optimizes the power flow distribution, and minimizes network losses.

1. Introduction

As modernization construction accelerates, industries in China experience rapid growth, leading to a steep rise in industrial power load and greater power supply demands. With the ongoing trend towards achieving "carbon peak and carbon neutrality goals'', traditional distribution network supply modes for industrial parks struggle to meet their power requirements (Zhao et al., 2021; Iqbal and Siddiqui, 2017). By incorporating multiple types of distributed resources, the innovative distribution network with VPP enhances traditional distribution networks, enabling the interconnection of various energy forms, improving energy utilization rates, and catering to the power needs of modern parks. Consequently, researching novel VPP-based optimization strategies for distribution networks is essential for ensuring the system's safe and economically efficient operation.

Numerous scholars have researched optimal operation strategies for distribution networks. Quan et al. (2023) employs a two-stage optimization model that accounts for distribution network uncertainty, effectively reducing network losses and enhancing voltage quality; however, this model does not consider system economics. Zhu et al. (2022) proposes a scheduling model based on deep reinforcement learning, which improves distribution network economy and mitigates power fluctuations, albeit disregarding new energy operation randomness. Xu et al. (2023) utilizes scenario analysis to simulate new energy randomness and considers the impact of lightning weather on distribution networks, albeit without considering the role of SOP in these networks. Wang et al. (2023b) constructs a two-stage optimization model to optimize the capacity of SOP, thereby improving system economics and reducing operating losses, but does not address new energy applications. Notably, the aforementioned studies do not explore the integration of flexible load (FL) clusters in distribution network optimization.

Effective dispatching of FL clusters can reduce system operating costs and stabilize voltage levels. Li et al. (2022b) constructs a model of electric vehicle (EV) FL clusters using a clustering algorithm; however, it does not consider cluster dispatching. Cedillo et al. (2022) establishes a dispatching model of the EV FL cluster with dynamic electricity pricing to enhance the revenue of charging stations, but does not account for the impact of the dispatching process on the distribution network. Xi et al. (2023) examines the influence of the EV FL cluster on distribution networks, yet only considers one type. Li et al. (2021) creates multiple dispatching models for FL clusters, including EV and air conditioning (AC), improving distribution network economy and security without considering the interconnection of other distributed energy sources. Building upon FL models, Chen et al. (2023) optimizes the location of distributed power supplies to enhance resilience against

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natural risks, but does not analyze the impact on system voltage quality. Notably, the studies mentioned above do not consider the effects of interconnection between VPP and distribution networks.

To address the limitations of existing research, this paper presents a bi-level optimization model for a novel distribution network based on the NNC method, incorporating VPP and FL clusters. The distribution network layer aims to minimize voltage offset and operation costs and obtains the Pareto front using the NNC method. It then selects the optimal solution via the TOPSIS method. The VPP optimization layer seeks to minimize operation costs, using the ATC algorithm to solve the bi-level model. The power flow model is transformed into a second-order cone programming model using second-order cone relaxation techniques. The uncertainty of new energy sources is addressed with chance-constrained programming, considering demand response (DR), FL clustering, SOP and SVC dispatching. The new energy sources chiefly comprise wind turbine generators (WTG) and photovoltaic generators (PVG), while the FL clusters mainly include EV and AC clusters. Finally, the improved IEEE 33-node system simulations are conducted to validate the effectiveness of the proposed model.

2. FL cluster model

2.1. EV cluster load model

In this paper, the primary focus of the EV cluster is private car users whose travel patterns are influenced by working hours. By analyzing statistical results from the Federal Highway Administration (Santos et al., 2011), we can obtain the improved probability density function representing the time and distance traveled by EV users between their work and residential areas. Consequently, the time probability density function for EV users arriving at the industrial park in the morning can be described as follows:

$$f\left(t_{\mathrm{r}}\right)=\begin{cases}\frac{1}{\sigma_{\mathrm{r}}\sqrt{2\Pi}}\exp\left[-\frac{\left(t_{\mathrm{r}}-\mu_{\mathrm{r}}\right)^2}{2\sigma_{\mathrm{r}}^2}\right], \mu_{\mathrm{r}}-12\leq t_{\mathrm{r}}<24\\ \frac{1}{\sigma_{\mathrm{r}}\sqrt{2\Pi}}\exp\left[-\frac{\left(t_{\mathrm{r}}-\mu_{\mathrm{r}}+24\right)^2}{2\sigma_{\mathrm{r}}^2}\right], 0\leq t_{\mathrm{r}}<\mu_{\mathrm{r}}-12\end{cases} \tag{1}$$

Where: t_{r} is the time when EV users arrive at the industrial park; $\mu_{\mathrm{r}}=8.8; \sigma_{\mathrm{r}}=2.55$.

The time probability density function for EV users departing from the industrial park in the afternoon can be described as follows:

$$f\left(t_{\mathrm{d}}\right)=\begin{cases}\frac{1}{\sigma_{\mathrm{d}}\sqrt{2\Pi}}\exp\left[-\frac{\left(t_{\mathrm{d}}-\mu_{\mathrm{d}}\right)^2}{2\sigma_{\mathrm{d}}^2}\right], & \mu_{\mathrm{d}}-12\leq t_{\mathrm{d}}<24\\ \frac{1}{\sigma_{\mathrm{d}}\sqrt{2\Pi}}\exp\left[-\frac{\left(t_{\mathrm{d}}-\mu_{\mathrm{d}}+24\right)^2}{2\sigma_{\mathrm{d}}^2}\right], & 0\leq t_{\mathrm{d}}<\mu_{\mathrm{d}}-12\end{cases} \tag{2}$$

Where: t_{d} is the time when the EV user leaves the industrial park; $\mu_{\mathrm{d}}=17.2; \sigma_{\mathrm{d}}=3.4$.

The probability density function representing the distance traveled by EV users from their residences to the industrial park can be described as follows:

$$f(l_s)=\frac{1}{s\sigma_s\sqrt{2\Pi}}\exp[-\frac{(\ln l_s-\mu_s)^2}{2\sigma_s^2}] \tag{3}$$

Where: l_{s} is the distance traveled by EV users from their residence to the industrial park; $\mu_{\mathrm{s}}=3.2; \sigma_{\mathrm{s}}=0.88$.

It is assumed that the EV begins charging upon the user's arrival at the industrial park, and the charging duration is as follows:

$$\begin{cases}T_{\mathrm{cha}}=\min(T_{\mathrm{soc}},T_{\mathrm{stay}})\\ T_{\mathrm{soc}}=\frac{l_{\mathrm{s}}E_{\mathrm{b}}}{l_{\max}p_{\mathrm{cha}}\eta_{\mathrm{cha}}}\\ T_{\mathrm{stay}}=t_{\mathrm{d}}-t_{\mathrm{r}}\end{cases} \tag{4}$$

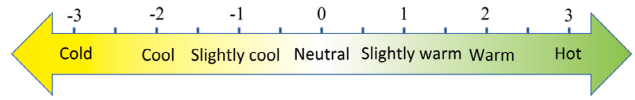

Fig. 1. Relationship between PMV value and human comfort.

Where: E_{b} and $l_{\max}$ are the battery capacity and the maximum travel distance of the EV respectively; p_{cha} and η_{cha} are the charging power and charging efficiency of the EV respectively.

Utilizing Eqs. (1)–(4), the charging power of an individual EV is obtained through Monte Carlo simulation (Wang and Infield, 2018). The total load for the EV cluster is subsequently determined by summing the power of all single EVs, as follows:

$$\begin{cases}P_{n,\mathrm{EV}}^t=p_{\mathrm{cha}}T_{\mathrm{cha}}^{n,t}\\ P_{\mathrm{EV}}^t=\sum_{n=1}^{n_{\mathrm{EV}}}P_{n,\mathrm{EV}}^t\end{cases} \tag{5}$$

Where: $P_{\mathrm{n,EV}}^t$ and P_{EV}^t are the charging power of the nth EV and EV cluster at time t obtained by Monte Carlo simulation, respectively; $T_{\mathrm{cha}}^{n,t}$ and n_{Ev} are the charging time of the nth EV and the number of EV at time t.

The following constraints must be met for EV clusters:

(1) Prevention of new peak constraint

$$P_{\mathrm{EVp}}^t\leq P_{\mathrm{EV}}^{\max} \tag{6}$$

Where: $P_{\mathrm{EV}}^{\max}$ and P_{EVp}^t are the maximumpower when EV cluster does not participate in dispatching and the power at time t after EV cluster participates in dispatching, respectively.

(2) Constraint of charging power conservation

$$\sum_{t=1}^{T_{\mathrm{cha}}}P_{\mathrm{EVp}}^t=\sum_{t=1}^{T_{\mathrm{cha}}}P_{\mathrm{EV}}^t \tag{7}$$

2.2. AC cluster load model

The thermal–mechanical model of the AC-building system can be characterized by equivalent parameters (Bao et al., 2023):

$$\frac{\mathrm{d}T_{\mathrm{in}}}{\mathrm{d}t}=\frac{T_{\mathrm{out}}-T_{\mathrm{in}}-Q_{\mathrm{AC}}R_{\mathrm{a}}}{R_{\mathrm{a}}C_{\mathrm{a}}} \tag{8}$$

Where: T_{in} and T_{out} are indoor and outdoor temperature; R_{a} and C_{a} are equivalent thermal resistance and heat capacity; Q_{AC} is the cooling (heat) of AC.

The electrical power consumed by AC cooling (heating) can be calculated by the energy efficiency ratio:

$$P_{\mathrm{AC}}=\frac{Q_{\mathrm{AC}}}{\gamma_{\mathrm{AC}}} \tag{9}$$

Where: P_{AC} and γ_{AC} are the electric power and energy $_{\mathrm{AC}}$ and γ_{AC} are efficiency ratio of AC consumption, respectively.

Assuming that the AC-building system maintains equilibrium, the AC cooling capacity and outdoor heat transfer become equal when AC electrical power and indoor temperature remain constant. By setting Eq. (8) equal to zero, the corresponding Eq. (9) can be derived as follows:

$$P_{\mathrm{AC}}=\frac{Q_{\mathrm{AC}}}{\gamma_{\mathrm{AC}}}=\frac{T_{\mathrm{out}}-T_{\mathrm{in}}}{R_{\mathrm{a}}\gamma_{\mathrm{AC}}} \tag{10}$$

From Eq. (10), it can be inferred that the AC electrical power level can be adjusted by altering the AC temperature. However, changes in indoor temperature can affect human comfort. The Predicted Mean Vote (PMV) is an index that reflects the degree of human thermal comfort (Yang et al., 2021a), as illustrated in Fig. 1.

Factors that influence human comfort encompass temperature, humidity, air flow rate, activity level, and clothing thickness, among others (Sunagawa et al., 2023). If we consider only the impact of temperature changes on PMV, assuming that other factors remain at optimal comfort levels, the relationship between PMV value and temperature can be derived as follows (Wang et al., 2023c):

$$\varphi_{\mathrm{PMV}}=\begin{cases}0.3895(T_{\mathrm{in}}-26), 26\le T_{\mathrm{in}}\\0.4065(T_{\mathrm{in}}-26), 26>T_{\mathrm{in}}\end{cases}\tag{11}$$

By combining Eqs. (10) and (11), the relationship between electrical power and AC PMV can be derived as follows:

$$P_{\mathrm{AC}}=\begin{cases}\dfrac{T_{\mathrm{out}}-(26+\varphi_{\mathrm{PMV}}/0.3895)}{R_{\mathrm{a}}\gamma_{\mathrm{AC}}}, & 0\le\varphi_{\mathrm{PMV}}\\\dfrac{T_{\mathrm{out}}-(26+\varphi_{\mathrm{PMV}}/0.4065)}{R_{\mathrm{a}}\gamma_{\mathrm{AC}}}, & 0>\varphi_{\mathrm{PMV}}\end{cases}\tag{12}$$

Hence, the electrical power magnitude of a single AC at varying comfort levels can be determined, and the overall electrical power of the AC cluster can be obtained by summing the power of each individual AC.

$$P_{\mathrm{ACA}}^{t}=\sum_{n=1}^{n_{\mathrm{AC}}}P_{n,\mathrm{AC}}^{t}\tag{13}$$

Where: $P_{n,\mathrm{AC}}^{t}$ and P_{ACA}^{t} are the nth single AC electric power and AC cluster electric power at time t respectively.

Once the AC cluster load model is established, the dispatching of AC clusters can be accomplished by adjusting the indoor thermal comfort levels.

3. Optimization model of distribution network layer

3.1. Objective function

The distribution network layer model aims to minimize the combined operating cost C^{O} and voltage offset ΔU_{sum} :

$$\begin{cases}\min\left(C^{\mathrm{o}},\Delta U_{\mathrm{sum}}\right)\\\Delta U_{\mathrm{sum}}=\displaystyle\sum_{t=1}^{24}\sum_{j=1}^{n_{\mathrm{sode}}}\left|\left(U_{j,t}^{*}\right)^{2}-1\right|\\C^{\mathrm{O}}=C_{\mathrm{PV}}+C_{\mathrm{WT}}+C_{\mathrm{B}}+C_{\mathrm{dr}}\end{cases}\tag{14}$$

Where: C_{PV} and C_{WT} are the operating costs of PVG and WTG, respectively; C_{B} and C_{dr} are the cost of power purchase in the distribution network and the compensation cost of customer participation in DR, respectively; $U_{j,t}^{*}$ and n_{node} are the voltage per unit value of node j at time t and the number of system nodes, respectively.

The various costs for C^{O} are as follows:

(1) Operating costs of new energy

$$\begin{cases}C_{PV}=\displaystyle\sum_{z=1}^{n_{PV}}\sum_{t=1}^{24}q^{PV}p_{z,t}^{PV}+\sum_{z=1}^{n_{PV}}\sum_{t=1}^{24}l^{PV}\left(p_{z,t}^{PV,\max}-p_{z,t}^{PV}\right)\\C_{WT}=\displaystyle\sum_{z=1}^{n_{WT}}\sum_{t=1}^{24}q^{WT}p_{z,t}^{WT}+\sum_{z=1}^{n_{WT}}\sum_{t=1}^{24}l^{WT}\left(p_{z,t}^{WT,\max}-p_{z,t}^{WT}\right)\end{cases}\tag{15}$$

Where: q^{PV} and q^{WT} are the per unit power generation cost of PVG and WTG respectively; $p_{z,t}^{\mathrm{PV}}$ and $p_{z,t}^{\mathrm{WT}}$ are the active power injected by the zth PVG and WTG respectively at time t; $p_{z,t}^{\mathrm{PV,max}}$ and $p_{z,t}^{\mathrm{WT,max}}$ are the maximum power generation capacity of the zth PVG and WTG respectively at time t; l^{PV} and l^{WT} are the per unit power disposal cost of PVG and WTG respectively.

(2) Power purchase cost of distribution network

$$C_{\mathrm{B}}=\sum_{t=1}^{24}q_{t}^{\mathrm{B}}\left(p_{\mathrm{b},t}^{\mathrm{grid}}+p_{\mathrm{b},t}^{\mathrm{vpp}}\right)\tag{16}$$

Where: q_{t}^{B} and $p_{\mathrm{b},t}^{\mathrm{grid}}$ are the electricity price at time t and the electricity purchased by the distribution network from the superior grid, respectively; $p_{\mathrm{b},t}^{\mathrm{vpp}}$ is the electricity purchased from VPP.

(3) Cost of DR

$$C_{\mathrm{dr}}=\sum_{t=1}^{24}r_{\mathrm{dr}}p_{\mathrm{dr}}^{t}\tag{17}$$

Where: r_{dr} is the compensation price per power unit for participating in incentive-based DR, taking the valley time tariff; p_{dr}^{t} is the load shedding amount at time t.

3.2. Constraints

(1) Power flow constraint of distribution network

The active and reactive power balance must be satisfied when the distribution network is in operation (Jain et al., 2019):

$$\begin{cases}I_{ij,t}^{2}=\frac{P_{ij,t}^{2}+Q_{ij,t}^{2}}{U_{i,t}^{2}}\\p_{j,t}=P_{ij,t}-r_{ij}I_{ij,t}^{2}-\sum_{k\in\omega(j)}P_{jk,t}\\q_{j,t}=Q_{ij,t}-x_{ij}I_{ij,t}^{2}-\sum_{k\in\omega(j)}Q_{jk,t}\\U_{j,t}^{2}=U_{i,t}^{2}-2\left(r_{ij}P_{ij,t}+x_{ij}Q_{ij,t}\right)+\left(r_{ij}^{2}+x_{ij}^{2}\right)I_{ij,t}^{2}\end{cases}\tag{18}$$

Where: $p_{j,t}$ and $q_{j,t}$ are the active and reactive power injected by node j at time t; $P_{ij,t}$ and $Q_{ij,t}$ are the active and reactive power of branch ij at time t; r_{ij} and x_{ij} are the resistance and reactance of the branch; $U_{i,t}$, $U_{j,t}$ are the node voltages of nodes i and j at time t; $I_{ij,t}$ and $\omega(j)$ are the current of branch ij at time t and the set of nodes at the end of the branch headed by j; $P_{jk,t}$ and $Q_{jk,t}$ are the active and reactive power of branch jk at time t.

(2) Security constraints of distribution network

$$\begin{cases}0\le I_{ij,t}^{2}\le I_{ij,\max}^{2}\\U_{\min}^{2}\le U_{j,t}^{2}\le U_{\max}^{2}\end{cases}\tag{19}$$

Where: $U_{\max}$ and $U_{\min}$ are the maximum and minimum node voltage of the distribution network respectively; $I_{ij,\max}$ is the maximum current allowed to pass through the branch ij.

(3) Operation constraints of SVC

$$q_{z,\min}^{\mathrm{svc}}\le q_{z,t}^{\mathrm{svc}}\le q_{z,\max}^{\mathrm{svc}}\tag{20}$$

Where: $q_{z,t}^{\mathrm{svc}}$ is the zth SVC reactive power compensation amount at t time; $q_{z,\max}^{\mathrm{svc}}$ and $q_{z,\min}^{\mathrm{svc}}$ are the maximum and minimum limits of zth SVC reactive power compensation, respectively.

(4) Uncertainty constraints of new energy

Uncertainty is inherent in new energy operations (Ai et al., 2016), and this can be characterized through the use of chance-constrained programming (Zhang et al., 2023):

$$\begin{cases}P\left\{p_{z,t}^{\mathrm{WT}}+\mu_{z,t}^{\mathrm{WT}}\le p_{z,t,\max}^{\mathrm{WT}}\right\}\ge\alpha_{1}\\P\left\{p_{z,t}^{\mathrm{PV}}+\mu_{z,t}^{\mathrm{PV}}\le p_{z,t,\max}^{\mathrm{PV}}\right\}\ge\alpha_{2}\\q_{z,t}^{\mathrm{WT}}=p_{z,t}^{\mathrm{WT}}\tan\varphi, p_{z,t}^{\mathrm{WT}}\ge 0\\q_{z,t}^{\mathrm{PV}}=p_{z,t}^{\mathrm{PV}}\tan\varphi, p_{z,t}^{\mathrm{PV}}\ge 0\end{cases}\tag{21}$$

Where: α_1 and α_2 are the probability of condition satisfaction, taken as 90%; $\mu_{z,t}^{\mathrm{WT}}$ and $\mu_{z,t}^{\mathrm{PV}}$ are the mean values of predicted deviation of zth PVG and WTG in time t, taken as 0; $p_{z,t,\max}^{\mathrm{WT}}$ and $p_{z,t,\max}^{\mathrm{PV}}$ are the maximum active power output of zth WTG and PVG in time t, respectively; $q_{z,t}^{\mathrm{WT}}$ and $q_{z,t}^{\mathrm{PV}}$ are the reactive power output of zth WTG and PVG in time t, respectively.

(5) Constraints of FL cluster

Power constraints of FL cluster:

The constraints for EV and AC clusters are discussed in Sections 2.1 and 2.2; therefore, they are not elaborated upon in this section.

Penetration constraint of FL cluster:

$$p_{\mathrm{f,max}}\ge\psi p_{\mathrm{dn,max}}\tag{22}$$

Where: $p_{\mathrm{f,max}}$ and $p_{\mathrm{dn,max}}$ are the maximum power in a day of FL cluster and distribution network respectively; ψ is the power penetration rate of FL cluster.

(6) SOP operation constraints

SOP primarily encompasses power and capacity constraints on both sides. Due to its minimal loss, this paper does not consider the loss aspect. The constraints can be expressed as follows (Deakin et al., 2023):

$$\begin{cases} P_{i,t}^{\mathrm{sop}} + P_{j,t}^{\mathrm{sop}} = 0 \\ \sqrt{\left(P_{j,t}^{\mathrm{sop}}\right)^2 + \left(Q_{j,t}^{\mathrm{sop}}\right)^2} \le S_j^{\mathrm{sop}} \\ \sqrt{\left(P_{i,t}^{\mathrm{sop}}\right)^2 + \left(Q_{i,t}^{\mathrm{sop}}\right)^2} \le S_i^{\mathrm{sop}} \end{cases} \tag{23}$$

Where: $P_{i,t}^{\mathrm{sop}}$ and $P_{j,t}^{\mathrm{sop}}$ are the output active power and injected active power of SOP at nodes i and j at time t, respectively; $Q_{i,t}^{\mathrm{sop}}$ and $Q_{j,t}^{\mathrm{sop}}$ are the output reactive power and injected reactive power of SOP at nodes i and j at time t, respectively; S_i^{sop} and S_j^{sop} are the capacities of SOP at nodes i and j at time t, respectively.

(7) Interactive power constraint

$$p_{\mathrm{b,min}}^{\mathrm{vpp}} \le p_{\mathrm{b,t}}^{\mathrm{vpp}} \le p_{\mathrm{b,max}}^{\mathrm{vpp}} \tag{24}$$

Where: $p_{\mathrm{b,max}}^{\mathrm{vpp}}$ and $p_{\mathrm{b,min}}^{\mathrm{vpp}}$ are the maximum and minimum electric power purchased from VPP, respectively.

(8) DR constraint

Price-based DR can alter the electricity consumption habits of certain customers, consequently adjusting their overall electricity usage (Yang et al., 2021b). It is crucial to ensure that these adjustments adhere to the corresponding constraints:

$$\begin{cases} P_{t,\mathrm{e}}^{\mathrm{DN}} = P_{t,\mathrm{load}}^{\mathrm{DN}} + D_{t,\mathrm{up}} - D_{t,\mathrm{down}} \\ \sum_{t=1}^{T} D_{t,\mathrm{up}} = \sum_{t=1}^{T} D_{t,\mathrm{down}} \\ 0 \le D_{t,\mathrm{up}} \le P_{t,\mathrm{load}}^{\mathrm{DN}} \zeta \\ 0 \le D_{t,\mathrm{down}} \le P_{t,\mathrm{load}}^{\mathrm{DN}} \zeta \end{cases} \tag{25}$$

Where: $P_{t,\mathrm{load}}^{\mathrm{DN}}$ and $P_{t,\mathrm{e}}^{\mathrm{DN}}$ are the electrical loads before and after the DR of the distribution network at time t, respectively; $D_{t,\mathrm{up}}$ and $D_{t,\mathrm{down}}$ denotes the amount of load increased and decreased by the price of electricity at time t, respectively; ζ denotes the maximum ratio of transferable load.

Incentive-based DR constraints:

$$0 \le D_{t,\mathrm{d}} \le P_{t,\mathrm{load}}^{\mathrm{DN}} \zeta \tag{26}$$

Where: $D_{t,\mathrm{d}}$ is the load reduction at time t.

3.3. Second-order cone model conversion

The deformation of the squared term for current and voltage can be represented as follows:

$$\begin{cases} \tilde{I}_{ij,t} = I_{ij,t}^2 \\ \tilde{U}_{j,t} = U_{j,t}^2 \end{cases} \tag{27}$$

By incorporating the above deformation into Eq. (18), the modified power flow constraint can be derived as follows:

$$\begin{cases} \tilde{I}_{ij,t}\tilde{U}_{i,t} = P_{ij,t}^2 + Q_{ij,t}^2 \\ p_{j,t} = P_{ij,t} - r_{ij}\tilde{I}_{ij,t} - \sum_{k \in \omega(j)} P_{jk,t} \\ q_{j,t} = Q_{ij,t} - x_{ij}\tilde{I}_{ij,t} - \sum_{k \in \omega(j)} Q_{jk,t} \\ \tilde{U}_{j,t} = \tilde{U}_{i,t} - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + (r_{ij}^2 + x_{ij}^2)\tilde{I}_{ij,t} \end{cases} \tag{28}$$

The security constraints for the distribution network can be adapted as follows:

$$\begin{cases} 0 \le \tilde{I}_{ij,t} \le I_{ij,\max}^2 \\ U_{\min}^2 \le \tilde{U}_{i,t} \le U_{\max}^2 \end{cases} \tag{29}$$

The second-order cone relaxation technique is applied to address Eqs. (23) and (28), resulting in the following (Li et al., 2022a):

$$\begin{cases} \left\| \begin{bmatrix} P_{i,t}^{\mathrm{sop}} & Q_{i,t}^{\mathrm{sop}} \end{bmatrix}^{\mathrm{T}} \right\| \le S_i^{\mathrm{sop}} \\ \left\| \begin{bmatrix} P_{j,t}^{\mathrm{sop}} & Q_{j,t}^{\mathrm{sop}} \end{bmatrix}^{\mathrm{T}} \right\| \le S_j^{\mathrm{sop}} \\ \left\| \begin{matrix} 2P_{ij,t} \\ 2Q_{ij,t} \\ \tilde{I}_{ij,t} - \tilde{U}_{i,t} \end{matrix} \right\| \le \tilde{I}_{ij,t} + \tilde{U}_{i,t} \end{cases} \tag{30}$$

Following the aforementioned derivation, the original flow model is transformed into a second-order cone programming problem.

4. Model of VPP optimization layer

4.1. Objective function

The primary objective of the VPP optimization layer is to minimize the overall cost.

$$\min C^{\mathrm{VPP}} = C_{\mathrm{fuel}} + C_{\mathrm{om}} + C_{\mathrm{idr}} - C_{\mathrm{se}} \tag{31}$$

Where: C_{fuel} and C_{om} are the fuel cost and operation cost of VPP respectively; C_{idr} and C_{se} are the integrated DR cost and income from electricity sales of VPP respectively.

The detailed computation of each cost component within the VPP is outlined below:

(1) Fuel cost for VPP

$$C_{\mathrm{fuel}} = \frac{\lambda_{\mathrm{gas}}}{q_{\mathrm{gas}}} \sum_{t=1}^{T} \left(\frac{P_{\mathrm{t,gt}}}{\eta_{\mathrm{ggt}}} + \frac{Q_{\mathrm{t,gb}}}{\eta_{\mathrm{gb}}} \right) \tag{32}$$

Where: λ_{gas} and q_{gas} denote the market price and heat value of natural gas, respectively; $P_{t,\mathrm{gt}}$ and η_{ggt} are the power and efficiency of gas turbine generation (GTG) at time t, respectively; $Q_{t,\mathrm{gb}}$ and η_{gb} are the power and thermal efficiency of gas boiler (GB) heating at time t.

(2) Operational cost for VPP

$$C_{\mathrm{om}} = \sum_{t=1}^{T} \Big[q^{\mathrm{PV}} P_{t,\mathrm{pv}} + q^{\mathrm{WT}} P_{t,\mathrm{wt}} + q_{\mathrm{ph}} \left(P_{t,\mathrm{ph}}^{\mathrm{d}} + P_{t,\mathrm{ph}}^{\mathrm{c}} \right) + q_{\mathrm{csp}} \left(Q_{t,\mathrm{sp}}^{\mathrm{d}} + Q_{t,\mathrm{cpp}}^{\mathrm{c}} \right) + q_{\mathrm{gt}} P_{t,\mathrm{gt}} + q_{\mathrm{re}} Q_{t,\mathrm{re}} + q_{\mathrm{ex}} Q_{t,\mathrm{ex}} + q_{\mathrm{gb}} Q_{t,\mathrm{gb}} + q_{\mathrm{eb}} Q_{t,\mathrm{eb}} + q_{\mathrm{ec}} Q_{t,\mathrm{ec}} + q_{\mathrm{ac}} Q_{t,\mathrm{ac}} \Big] \tag{33}$$

Where: $P_{t,\mathrm{pv}}$ and $P_{t,\mathrm{wt}}$ are the power of PVG and WTG of VPP at time t,respectively; q_{ph} and q_{csp} are the per unit power operation cost of pumped storage (PS) power station and concentrated solar power (CSP) station, respectively; $P_{t,\mathrm{ph}}^{\mathrm{d}}$ and $P_{t,\mathrm{ph}}^{\mathrm{c}}$ are the power output and power consumption of PS power station at time t, respectively; $Q_{t,\mathrm{csp}}^{\mathrm{d}}$ and $Q_{t,\mathrm{csp}}^{\mathrm{c}}$ are the heat release power and heat storage power of CSP station at time t, respectively; q_{gt} and q_{re} are the per unit power operation cost of GTG and waste heat boiler, respectively; q_{ex} and q_{gb} are the per unit power operating costs of heat exchangers (HE) and GB, respectively; $Q_{t,\mathrm{ex}}$ and $Q_{\mathrm{t,gb}}$ are the per unit power operating costs of HE and GB at time t,respectively; q_{eb} and q_{ec} are the per unit power operating costs of electric boilers (EB) and electric chillers (EC), respectively; $Q_{t,\mathrm{eb}}$ and $Q_{t,\mathrm{ec}}$ are the per unit power operating costs of EB and EC at time t respectively; q_{ac} and $Q_{t,\mathrm{ac}}$ are the per unit operating costs and cooling power of absorption chiller (AC) at time t,respectively; $Q_{\mathrm{t,re}}$ is the heating power of waste heat boiler at t time.

(3) Integrated DR operating costs

The cost compensation calculation method for the DR involving cooling, heating, and power loads within the VPP is identical to the DR of the power load at the distribution network layer. Therefore, it will not be elaborated upon further in this section.

(4) Electricity sales revenue

$$C_{se}=\sum_{t=1}^{T}q_t^{B}P_{se,t}^{DN} \tag{34}$$

Where: $P_{se,t}^{DN}$ is the electrical power sold to the distribution network by VPP at time t.

4.2. Constraints

(1) Energy balance constraint

Electrical balance constraint (Yang et al., 2023):

$$P_{t,load}^{vpp}=P_{t,pv}+P_{t,wt}+P_{t,ph}^{d}-P_{t,ph}^{c}+P_{t,csp}^{d}+P_{t,gt}-P_{t,eb}-P_{t,ec}-P_{b,t}^{DN} \tag{35}$$

Where: $P_{t,load}^{vpp}$ and $P_{t,eb}$ are the electric load and the electric power consumed by the EB after VPP DR at time t, respectively; $P_{t,csp}^{d}$ and $P_{t,ec}$ are the power generated by the CSP station and the electric power consumed by the electric chiller at time t, respectively.

Heat balance constraint (Yang et al., 2022):

$$Q_{t,hload}^{vpp}=Q_{t,ex}^{out}+Q_{t,eb}+Q_{t,gb} \tag{36}$$

Cooling balance constraint (Wang et al., 2023a):

$$Q_{t,cload}^{vpp}=Q_{t,ac}^{out}+Q_{t,ec} \tag{37}$$

Where: $Q_{t,cload}^{vpp}$ and $Q_{t,ac}^{out}$ are the cooling load and the power output of the AC after the DR of VPP at the time t, respectively.

(2) Operational constraints of PS power station

Reservoir capacity constraint (Gao et al., 2022):

$$\begin{cases} W_{t+1}^{ph}=W_t^{ph}+Q_{t,c}^{ph}-Q_{t,\,d}^{ph}+Q_{t,rain}^{ph} \\ Q_{t,c}^{ph}=\dfrac{3600\times1000\eta_c^{ph}P_{t,ph}^{ch}}{\rho gH} \\ Q_{t,\,d}^{ph}=\dfrac{3600\times1000P_{t,ph}^{d}}{\eta_d^{ph}\rho gH} \\ W_{min}^{ph}\le W_t^{ph}\le W_{max}^{ph} \end{cases} \tag{38}$$

Where: W_t^{ph} and W_{t+1}^{ph} represent the residual water in the reservoir at time t and $t+1$, respectively; $Q_{t,c}^{ph}$, $Q_{t,d}^{ph}$ represent the water flow in pumping and power generation mode at time t,respectively; η_c^{ph} and η_d^{ph} generation mode at time t, respectively; η_c^{ph} and η_d^{ph} are the efficiencies of pumping and power generation in PS power station, respectively; ρ and g represent the density of water and gravitational acceleration, respectively; W_{max}^{ph} and W_{min}^{ph} are the maximum andminimum limits of reservoir capacity, respectively; H represents the water head height; $Q_{t,rain}^{ph}$ is the water flow into the reservoir due to rainfall at time t, which is not considered in this paper.

Operational state constraints of PS power station:

$$\begin{cases} u_{t,ph}^{c}P_{ph,min}^{c}\le P_{t,ph}^{c}\le u_{t,ph}^{c}P_{ph,max}^{c} \\ u_{t,ph}^{d}P_{ph,min}^{d}\le P_{t,ph}^{d}\le u_{t,ph}^{d}P_{ph,max}^{d} \\ u_{t,ph}^{c}+u_{t,ph}^{d}\le 1 \end{cases} \tag{39}$$

Where: $u_{t,ph}^{c}$ and $u_{t,ph}^{d}$ are the pumping and power generation states of the PS power station at time t, respectively, as the Boolean variables; $P_{ph,max}^{c}$ and $P_{ph,min}^{c}$ are the maximum and minimum limits of pumping power consumption of the PS power station, respectively; $P_{ph,max}^{d}$ and $P_{ph,min}^{d}$ are the maximum and minimum limits of power generation of the PS station, respectively.

(3) Constraints in the operation of CSP station

Thermal energy flow constraints of heat transfer medium in CSP stations:

$$Q_{t,csp}^{d}=Q_{t,csp}^{hload}+Q_{t,csp}^{load} \tag{40}$$

Where: $Q_{t,csp}^{hload}$ and $Q_{t,csp}^{load}$ represents the heat release power of the CSP station used to satisfy the thermal and electrical loads, respectively.

Thermal storage capacity constraints for CSP station:

$$\begin{cases} W_{t+1}^{csp}=W_t^{csp}+Q_{t,csp}^{c}-Q_{t,csp}^{d} \\ W_{min}^{csp}\le W_t^{csp}\le W_{max}^{csp} \end{cases} \tag{41}$$

Where: W_t^{csp} and W_{t+1}^{csp} represents the heat stored in the CSP station at time and time $t+1$, respectively; W_{max}^{csp} and W_{nin}^{csp} represents the maximum and minimum limits of heat that can be stored in the CSP station,respectively.

Storage operation constraints of CSP station:

$$\begin{cases} u_{t,csp}^{c}P_{csp,min}^{c}\le Q_{t,csp}^{c}\le u_{t,csp}^{c}P_{csp,max}^{c} \\ u_{t,csp}^{d}P_{csp,min}^{d}\le Q_{t,csp}^{d}\le u_{t,csp}^{d}P_{csp,max}^{d} \\ u_{t,csp}^{c}+u_{t,csp}^{d}\le 1 \end{cases} \tag{42}$$

Where: $u_{t,csp}^{c}$ and $u_{t,csp}^{d}$ are the thermal storage and discharge states of the CSP station at time t respectively; $P_{csp,max}^{c}$ and $P_{csp,min}^{c}$ represent the maximum and minimum limits of the thermal storage power of the CSP station respectively; $P_{csp,max}^{d}$ and $P_{csp,min}^{d}$ represent the maximum and minimum limits of the discharge power of the CSP station respectively.

Power generation constraints of CSP station:

$$P_{t,csp}^{d}=\eta_{csp}Q_{t,csp}^{load} \tag{43}$$

Where: η_{csp} represents the heat-to-electricity ratio of the CSP station.

(4) GTG constraint

GTG power constraints (Wang et al., 2023a):

$$\begin{cases} P_{gt,min}\le P_{t,gt}\le P_{gt,max} \\ r_{down}^{gt}\le P_{t,gt}-P_{t-1,gt}\le r_{up}^{gt} \end{cases} \tag{44}$$

Where: $P_{gt,max}$ and $P_{gt,min}$ are the maximum and minimum power generation limits of the GTG respectively; r_{up}^{gt} and r_{down}^{gt} are the up and down climbing rates of the GTG respectively.

The heat production constraint of the GTG is:

$$Q_{t,gt}=\eta_{gt}P_{t,gt} \tag{45}$$

Where: η_{gt} and $Q_{t,gt}$ represent the and $Q_{t,gt}$ represent the heat-to-electricity ratio of the GTG and the heat production power at time t, respectively.

5)GB constraint

$$Q_{gb,min}\le Q_{t,gb}\le Q_{gb,max} \tag{46}$$

Where: $Q_{gb,max}$ and $Q_{gb,min}$ are the maximum and minimum limits of the heating power of the GB respectively.

(6) Constraints of EB and EC

Constraints of EB operation:

$$\begin{cases} Q_{t,eb}=\eta_{eb}P_{t,eb} \\ P_{eb,min}\le P_{t,eb}\le P_{eb,max} \end{cases} \tag{47}$$

Where : $P_{eb,max}$ and $P_{eb,min}$ are the maximum and minimum limits of electric power for EB respectively; η_{eb} is the energy consumption ratio of EB.

Constraints of electric chiller operation:

$$\begin{cases} Q_{t,ec}=\eta_{ec}P_{t,ec} \\ P_{ec,min}\le P_{t,ec}\le P_{ec,max} \end{cases} \tag{48}$$

Table 1
Pareto front anchor point.

Objectives	f_1	f_2
$\min f_1(x)$	z_{11}	z_{12}
$\min f_2(x)$	z_{21}	z_{22}

Where : $P_{\text{eb,max}}$ and $P_{\text{eb,min}}$ are the maximum and minimum limits of power consumption of electric refrigeration machine; η_{eb} is the energy consumption ratio of electric chiller.

(7) Constraints of heat recovery boiler and heat exchanger

Operating constraints of heat recovery boiler:

$$Q_{t,\text{re}}^{\text{out}} = \eta_{\text{re}} Q_{t,\text{re}} \tag{49}$$

Where: η_{re} and $Q_{t,\text{re}}^{\text{out}}$ are the thermal efficiency of the heat recovery boiler and the output thermal power at time t, respectively.

Operating constraints of heat exchanger:

$$Q_{t,\text{ex}}^{\text{out}} = \eta_{\text{ex}} Q_{t,\text{ex}} \tag{50}$$

Where : η_{ex} and $Q_{t,\text{ex}}$ are the heat exchanger thermal efficiency and the input thermal power at time t, respectively.

(8) Constraint of AC

$$Q_{t,\text{ac}}^{\text{out}} = \eta_{\text{ac}} Q_{t,\text{ac}} \tag{51}$$

Where: η_{ac} and $Q_{t,\text{ac}}$ are the refrigeration efficiency of AC and input power at time t, respectively.

(9) Constraints of new energy

The operational constraints for new energy sources have already been introduced at the distribution network layer and will not be reiterated in this section.

(10) Interactive power constraints

The interactive power operation constraint has been introduced at the distribution network layer, which is not repeated here.

(11) Constraints of integrated DR

The constraints concerning the participation of cooling, heating, and electrical loads in the VPP's DR are identical to those for the DR at the distribution network layer. Therefore, they will not be detailed further in this section.

5. Method for solving the bi-level model

5.1. NNC method for solving multi-objective problems at the distribution network layer

The original optimization model for the distribution network layer can be expressed as follows:

$$\begin{cases} \min f_1(x) \\ \min f_2(x) \\ s.t \quad g(x) \le 0 \\ \qquad h(x) = 0 \end{cases} \tag{52}$$

Where: f_1 and f_2 are the objectives to be optimized; x is the variable to be optimized; $g(x)$ and $h(x)$ are the inequality and equation constraints, respectively.

The anchor points of the Pareto front can be acquired by independently optimizing each objective (Lahri et al., 2021). The results are displayed in Table 1.

Based on the data from the table, the anchor point coordinates are (z_{11}, z_{12}) and (z_{21}, z_{22}). The Pareto solution space is subsequently normalized as follows (Hashemi et al., 2023):

$$f = \left(\overline{f_1}, \overline{f_2}\right) = \left(\frac{f_1 - z_{11}}{z_{21} - z_{11}}, \frac{f_2 - z_{22}}{z_{12} - z_{22}}\right) \tag{53}$$

Following the normalization process, the coordinates of the solution set anchor points transform into $A_1(0,1)$ and $A_2(1,0)$,a new anchor point linkage is defined as the utopia line A. On the utopia line A, the vector pointing from A_1 to A_2 is $\vec{A} = \sqrt{2}\angle -45°$, and dividing the line A into n equal parts, which will produce $n+1$ equidistant splitting points. The coordinates of zth splitting point A_z can be expressed as follows: be expressed as follows:

$$A_z = \left(1 - \frac{z}{n}\right) A_1 + \frac{z}{n} A_2 \text{ , } z = 0, 1, \dots, n \tag{54}$$

Upon determining the splitting point coordinates, a vertical line intersecting the utopia line is drawn at the splitting point A_z. This vertical line intersects the Pareto front at the point B_z. The point B_z is the optimal solution of the corresponding splitting point. Now define $\overline{A_z}$ and $\overline{f}$ as vectors with the origin pointing to A_z and f,respectively, from which the new constraint can be obtained as follows:

$$\vec{A}\left(\vec{f} - \overline{A_z}\right) \le 0 \tag{55}$$

The optimal solution B_z corresponding to the splitting point A_z is attained from Eq. (55). Therefore, Eq. (52) can be restructured as follows:

$$\begin{cases} \min f_2(x) \\ s.t \ g(x) \le 0 \\ h(x) = 0 \\ \overline{A}\left(\overline{f} - \overline{A_z}\right) \le 0 \end{cases} \tag{56}$$

Based on the aforementioned analysis, each splitting point is sequentially substituted into $\overline{A_z}$ to derive the entire Pareto solution set. Ultimately, the balanced solution can be chosen using the TOPSIS method (Xu et al., 2022), which is then passed to the VPP layer.

5.2. Solve bi-level model with ATC algorithm

The ATC algorithm, suitable for solving bi-level models, enhances the solving speed and achieves iterative convergence (Wang et al., 2021). In this study, the power of the tie line between the distribution network and VPP is considered as the virtual source for the distribution network and the virtual load for VPP. Consequently, the Lagrangian function is constructed as follows:

(1) The objective for the distribution network layer can be reformulated as follows:

$$\min f_2(x) + \sum_{t=1}^{T} \omega_t \left(\overline{P_{\text{se},t}^{\text{DN}}} - p_{\text{b},t}^{\text{vpp}}\right) + \sum_{t=1}^{T} \gamma_t \left(\overline{P_{\text{se},t}^{\text{DN}}} - p_{\text{b},t}^{\text{vpp}}\right)^2 \tag{57}$$

Where: ω_t and γ_t represent the Lagrange multipliers at time t; $\overline{P_{\text{se},t}^{\text{DN}}}$ is the coupling variable obtained from the VPP at time t.

(2) The objective of the VPP optimization layer can be reformulated as follows:

$$\min C^{\text{VPP}} + \sum_{t=1}^{T} \omega_t \left(P_{\text{se},t}^{\text{DN}} - \overline{p_{\text{b},t}^{\text{vpp}}}\right) + \sum_{t=1}^{T} \gamma_t \left(P_{\text{se},t}^{\text{DN}} - \overline{p_{\text{b},t}^{\text{vpp}}}\right)^2 \tag{58}$$

Where: $\overline{p_{\text{b},t}^{\text{vpp}}}$ is the coupling variable obtained from the distribution network at time t.

The bi-level solving process based on the ATC algorithm is as follows:

Step 1: Initialize the Lagrange multiplier and set the number of iterations k=1.

Step 2: Solve the distribution network optimization layer and pass the value of $\overline{p_{\text{b},t}^{\text{vpp}}}$ for each hour to VPP.

Step 3: After obtaining the distribution network data, the VPP optimization layer model is solved and the value of $\overline{P_{\text{se},t}^{\text{DN}}}$ for each hour is passed to the distribution network.

$$\left| p_{\text{b},t}^{\text{vpp},k} - P_{\text{se},t}^{\text{DN},k} \right| \le \varepsilon \tag{59}$$

Table 2
Results of model simulation.

Items	Results
Cost of VPP	658823.89(RMB)
Cost of distribution network	34806.02(RMB)
Loss of distribution network	2190.07 (kW)
Voltage offset	44.60 (p.u.)

Step 5: Update the Lagrange multiplier according to Eq. (60), $k = k+1$, and go back to Step 2.

$$\begin{cases} \omega_t^{k+1} = \omega_t^k + 2\left(\gamma_t^k\right)^2 \left(p_{\mathrm{b},t}^{\mathrm{vpp},k} - P_{\mathrm{se},t}^{\mathrm{DN},k}\right) \\ \gamma_t^{k+1} = \beta \gamma_t^k \end{cases} \tag{60}$$

The flow chart is shown in Fig. A.1 in Appendix A.

6. Case study

In this study, the example is validated using an improved IEEE 33-node distribution system, as illustrated in Fig. A.2 in Appendix A. There are 200 EVs installed at nodes 8 and 32, and 200 ACs installed at nodes 21 and 24. A tie line is constructed at node 16 to connect with the VPP, with a maximum power capacity of 500 kW. WTG with a capacity of 400 kW is installed at nodes 4 and 14, while PVG with a capacity of 400 kW are installed at nodes 11 and 29; The SOP with a capacity of 100 kVA is installed between nodes 25 and 33, and the SVC with a capacity of 400 kvar is installed at nodes 18 and 33. The load and parameters of each piece of equipment are depicted in Figs. A.3–A.5 in Appendix A and Table B.1 in Appendix B. The initial value of the Lagrange multiplier is set to 1, the convergence accuracy is set to 0.001, and the penetration of the FL cluster is 20%. The model in this study is solved using CPLEX in MATLAB 2018.

6.1. Analysis of model results in this paper

The simulation results of the model presented in this paper can be found in Table 2, as well as Figs. 2–5.

Table 2 clearly demonstrates that the operation cost generated by the VPP is significantly higher than the distribution network's operation cost, due to its substantial load.

As depicted in Fig. 2, the introduction of second-order cone relaxation may result in errors in the calculation. However, these errors are of the order of 10^{-4} and less than 2.5×10^{-4}, ensuring an acceptable level of accuracy.

Fig. 3 illustrates that DR can effectively reduce the electrical load of the VPP during peak hours, thereby alleviating the pressure on its power supply. Furthermore, the power sold by the VPP to the distribution network can provide an active power source for the network. This not only avoids any negative impact on the VPP but also generates some revenue, as the proportion of power sold is relatively small.

Fig. 4 indicates that DR effectively achieves peak-shaving for heat load. Among the heating equipment, it is apparent that the heat exchanger serves as the primary source of heat energy. By employing electric and GB as supplementary heating equipment, the overall energy utilization is enhanced.

6.2. Analysis of case comparison

To verify the effective of the optimization model proposed in this paper, the results of the following cases are compared and analyzed respectively.

(1) Case 1: The optimization model proposed in this paper.

(2) Case 2: Using Case 1 as a foundation, the FL cluster is not included in the optimal dispatching process.

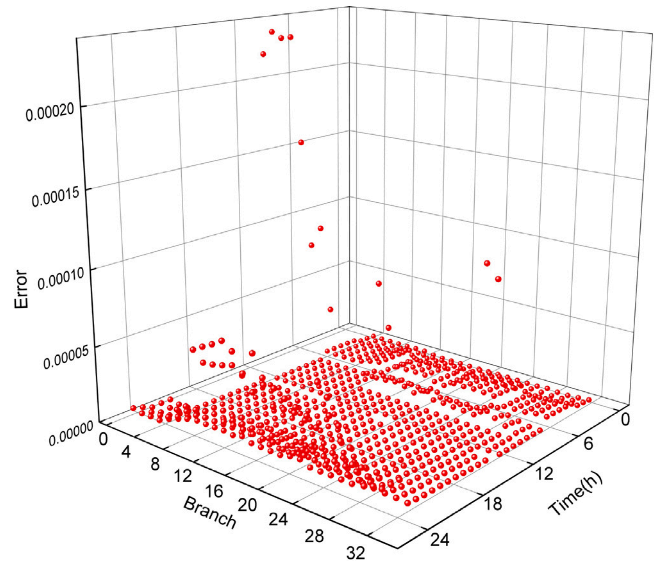

Fig. 2. Relaxation error.

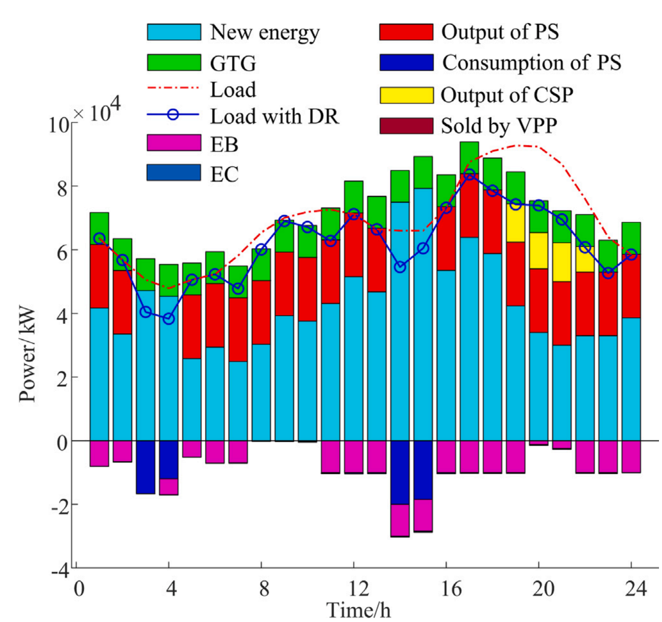

Fig. 3. Electrical load balance curve of VPP.

(3) Case 3: Based on Case 2, the SOP is not taken into account.

(4) Case 4: Based upon Case 3, the interconnection between the VPP and the distribution network is not taken into account.

The optimization results for each case are presented in Table 3 and illustrated in Figs. 6–10.

Table 3 presents a comparison of the results between Cases 3 and 4, demonstrating that when the VPP is interconnected with the distribution network via the tie line, the operation cost of the distribution network is reduced by 6.31%, the voltage offset increases by 3.23 p.u., and the operation cost of the VPP is decreased by 1349.17RMB due to the distribution network's power purchases from the VPP. Consequently, the interconnection of the VPP with the distribution network significantly reduces both the distribution network's and the VPP's operation costs, although it leads to an increase in the voltage offset. Upon comparing Cases 2 and 3, we observe that after the inclusion of SOP in the distribution network, it improves the power flow, reduces the voltage offset from 42.77 to 42.02p.u., and decreases the operation

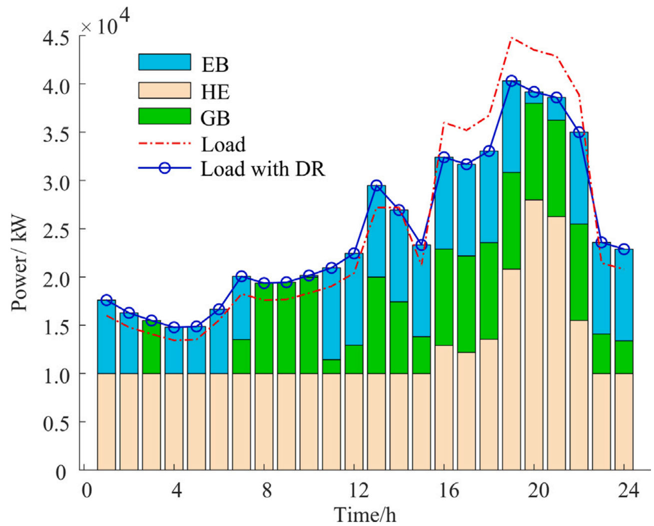

Fig. 4. Heat load balance curve of VPP.

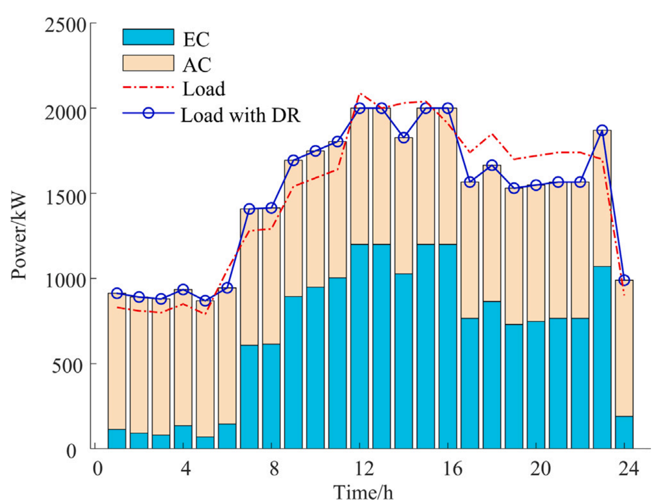

Fig. 5. Cooling load balance curve of VPP.

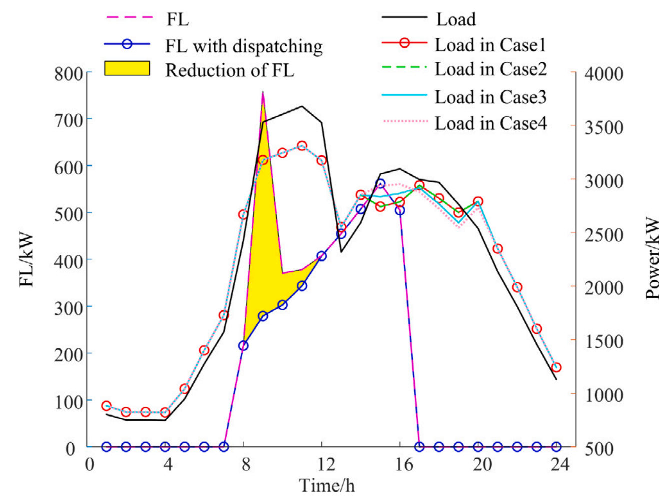

Fig. 6. Load optimization results of distribution network.

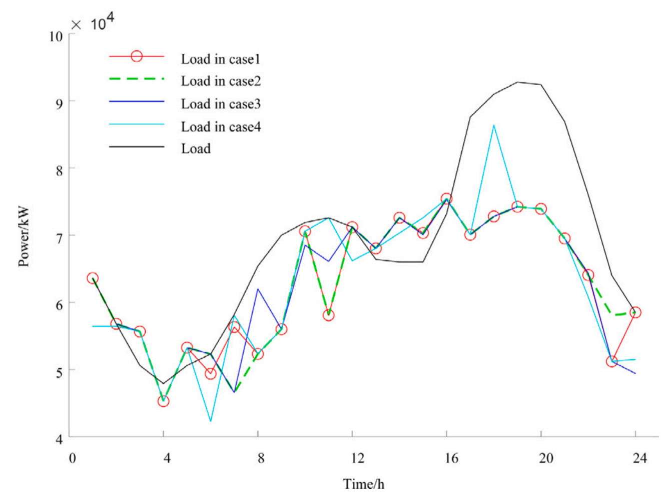

Fig. 7. Electrical load optimization results of VPP.

Table 3
Comparison of strategies.

Cases	Cost of distribution network /RMB	Voltage offset /p.u.	Cost of VPP /RMB
Case 1	34 806.02	44.60	658 823.89
Case 2	37 417.92	42.02	658 823.89
Case 3	40 345.81	42.77	657 474.76
Case 4	42 893.61	39.54	658 823.93

cost by 2927.89RMB. Although the operation cost of the VPP increases by 1349.13RMB, the total system cost still reduces by 1400.69RMB. Consequently, the addition of SOP to the distribution network evidently enhances voltage quality and lowers the overall system cost. Comparing Cases 1 and 2, it is evident that when the FL cluster is dispatched, the voltage offset in the distribution network increases 2.58p.u., and the operation cost decreases by 2611.9RMB. Despite these changes, the operation cost of the VPP remains unchanged. The involvement of the FL cluster in dispatching effectively reduces the operation cost of the distribution network. Hence, when both SOP and VPP are interconnected and the FL cluster is dispatched, it not only satisfies customers' power quality requirements but also decreases the system operation cost, optimizing the overall performance of the distribution network.

For the four cases, variations in the electrical load of the distribution network and VPP are illustrated in Figs. 6–10.

Fig. 6 reveals that when FL clusters in industrial areas participate in dispatching, power consumption of these clusters decreases between 9:00 and 12:00, which coincides with the peak power consumption period of industrial loads. Thus, the participation of FL s in dispatching exhibits a peak-shaving effect. Comparing the electric load of the distribution network under the four cases with the initial load, we observe that during the off-peak hours of 1:00–7:00, the electric load in all four cases is much higher than the initial load. Conversely, during peak hours of 9:00–12:00 and 15:00–18:00, the electric load under these four cases is lower than the initial load. During other periods, the load in all four cases experiences a slight increase.

Fig. 7 shows that during peak hours for the VPP, a portion of the load is either reduced or shifted to other time periods, and the electric load under all four strategies is also influenced by DR. Consequently, the implementation of DR at the customer side facilitates load shifting or reduction, thereby achieving peak shaving and valley filling effects.

When the distribution network load corresponds to its minimum and maximum values at 4:00 and 11:00, respectively, the voltage distribution of the distribution network for the four strategies is depicted in Figs. 8–9.

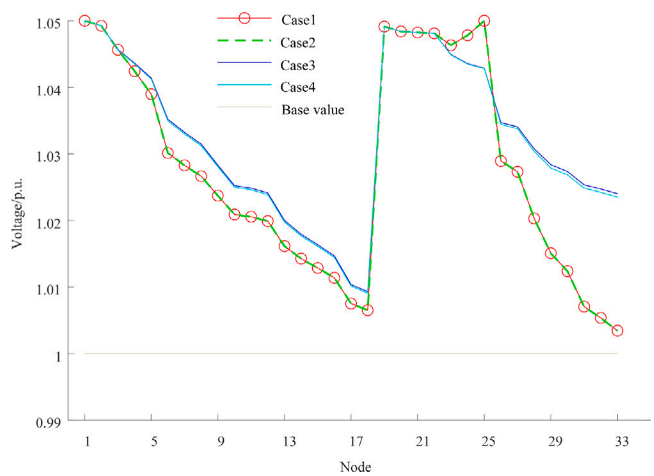

Fig. 8. Voltage of distribution network with minimum load.

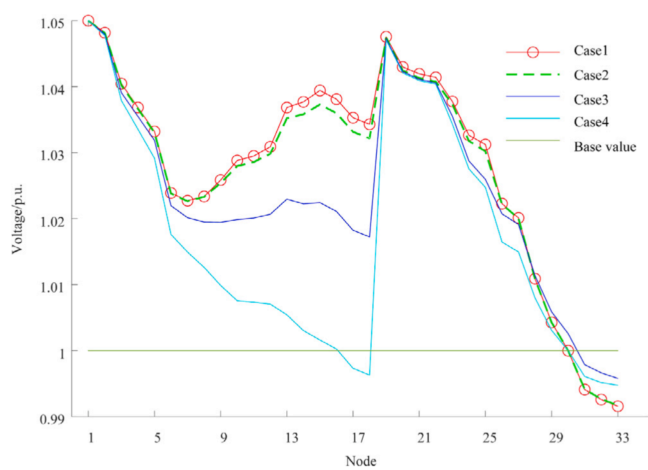

Fig. 9. Voltage of distribution network with maximum load.

As depicted in Fig. 8, the voltage levels at each node of the distribution network across all four cases exceed the base value when the network is operating at its minimum load. Notably, the voltage distribution is similar under both Case 1 and Case 2 operation modes, as well as under Case 3 and Case 4 operation modes. The voltage offset in the distribution network for Case 1 and Case 2 is smaller than that in Case 3 and Case 4, which can be attributed to the SOP.

As illustrated in Fig. 9, when the distribution network operates at maximum load, the voltage of the entire system experiences a drop due to excessive load in the absence of VPP connection and without considering the SOP, as demonstrated by the results of Case 4. Conversely, when connected to the VPP, the active power of the entire system is supplemented, resulting in an improved system voltage, as evident from the results of Case 3. Upon further connection of the SOP, the system's power flow is improved, and the voltage increases, as depicted in the results of Case 2. Additionally, when FL is dispatched during peak hours, the system's active load amount can be reduced, resulting in the largest average voltage magnitude, as demonstrated by the results of Case 1.

Fig. 10 presents the distribution of network loss for the distribution network under each case. It is evident from the figure that when operating under Case 4, the distribution network maintains a lower voltage level, and even though the voltage offset is smaller, it results in a higher network loss. When operating under Case 3, the distribution network is connected to the VPP, receiving active power, which in

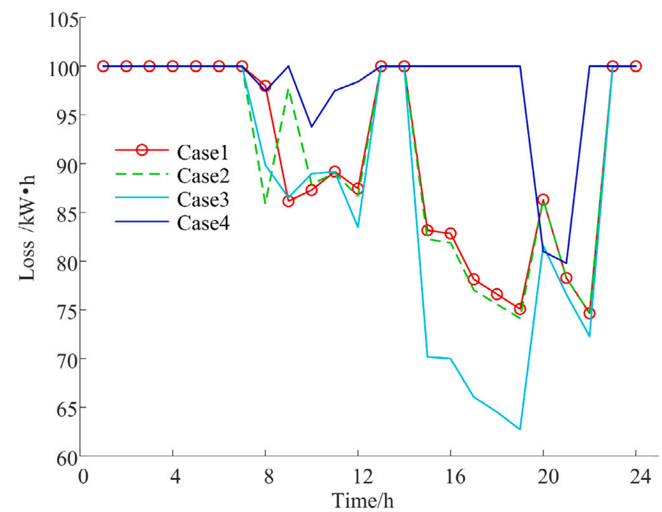

Fig. 10. Network loss of distribution network.

turn increases the system voltage amplitude and subsequently reduces network loss. Under Case 2 and Case 1, the distribution network is equipped with an SOP, enhancing the distribution of power flow. This combines the advantages of Case 3 and Case 4 while simultaneously considering both system network loss and voltage offset.

6.3. Consider the effect of FL penetration and AC comfort

The values of FL penetration and AC comfort influence the magnitude of FL, which in turn has specific effects on the operation of the distribution network and VPP. The system is simulated under varying conditions, incorporating 10%, 30%, and 50% of FL penetration levels and −1, 0, and 1 for AC PMV, respectively. The results are represented in Figs. 11–14.

As illustrated in Fig. 11, when the FL penetration rate remains constant, a smaller AC PMV value results in a larger difference between indoor and outdoor temperatures, leading to higher system electricity consumption and a higher total cost. Conversely, when the AC PMV remains constant and the FL penetration rate is greater, the amount of FL within the system increases, which directly raises the electric load and leads to a higher total cost.

As observed from Fig. 12 through Fig. 14, when the penetration rate of FL exceeds 50%, the change in penetration rate results in a smaller impact on voltage offset but causes a more significant alteration in network loss and distribution network operation cost, which increases alongside the penetration rate. When the FL penetration rate is less than 50%, an increase in penetration leads to a drop in system voltage, enabling the voltage at each node to approach the base value. This reduces the voltage offset while simultaneously increasing the network loss and operating costs of the distribution network. When the AC comfort value is greater than or equal to zero, the operating cost of the VPP remains unaffected by changes in FL penetration and maintains a stable constant value. However, it does result in a significant alteration in the operating cost and network loss of the distribution network. Consequently, appropriate dispatch of FL can achieve both economic efficiency and system stability.

6.4. Comparison of algorithms

To validate the efficacy of the NNC multi-objective algorithm presented in this paper, the algorithm's results are compared with those obtained using the ε-constrained method and the NSGA-II algorithm, as illustrated in Fig. 15 and Table 4.

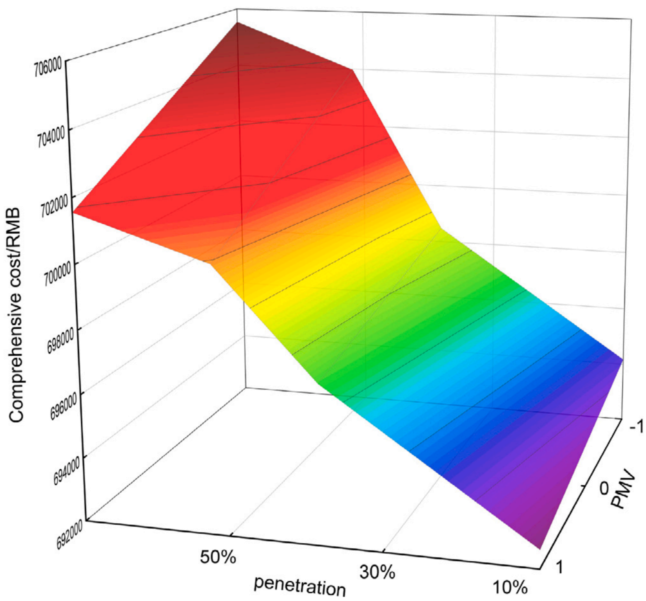

Fig. 11. Cost distribution.

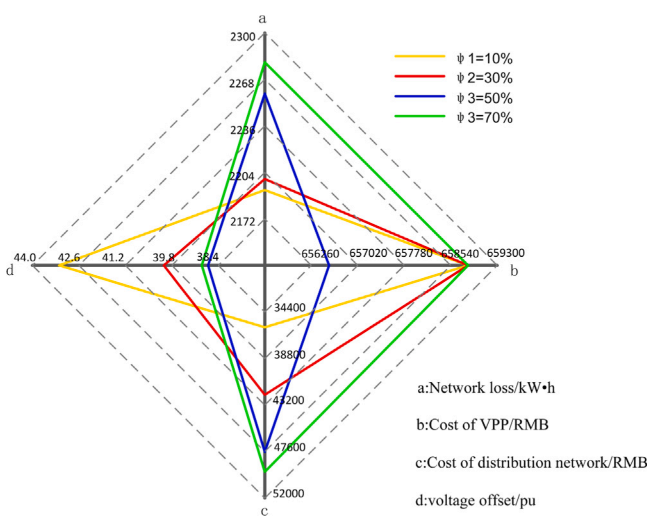

Fig. 12. Optimization results when PMV is -1.

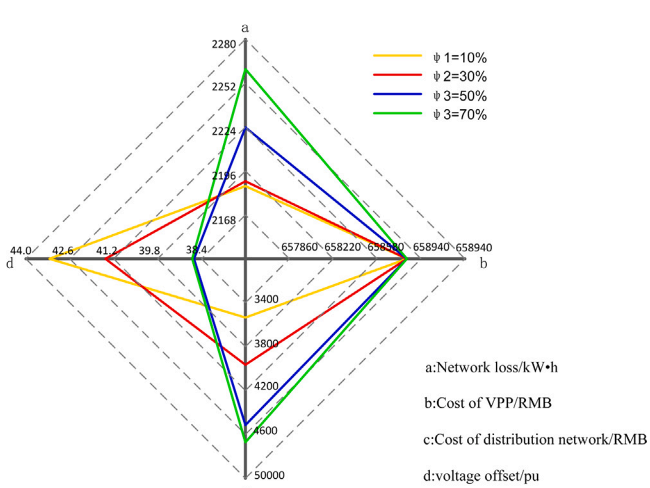

Fig. 13. Optimization results when PMV is 0.

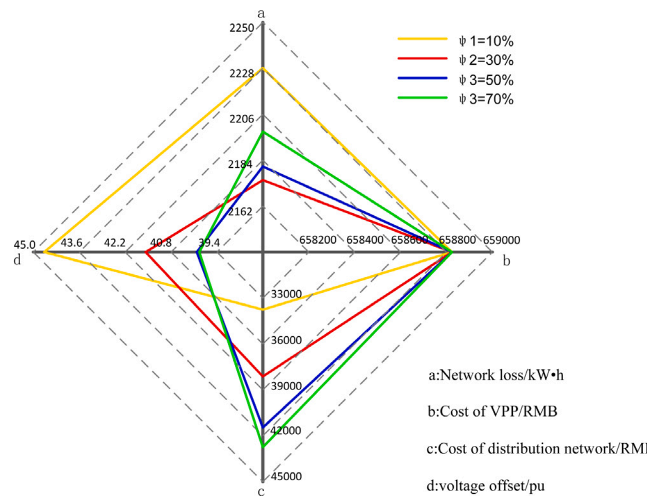

Fig. 14. Optimization results when PMV is 1.

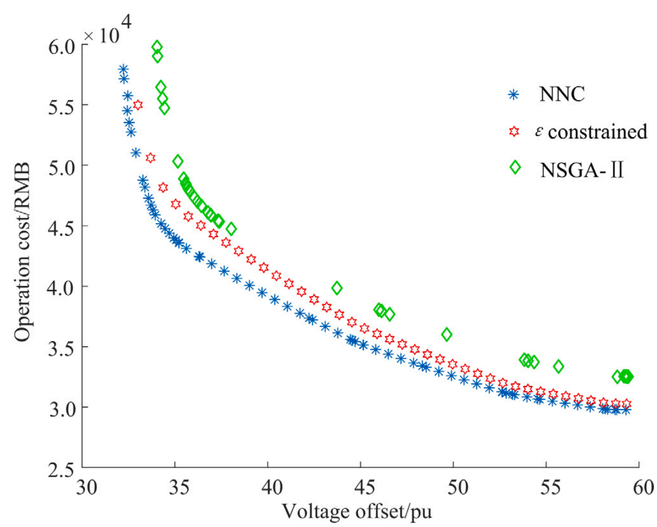

Fig. 15. Comparison of Pareto front.

As depicted in Fig. 15, when addressing multivariate problems, the NSGA-II algorithm is prone to getting trapped in a local optimum and fails to produce uniform Pareto fronts. While the ε-constrained method can obtain relatively uniform solutions overall in solving multi-objective problems, the solutions surrounding the anchor point are not uniformly distributed. The NNC multi-objective optimization algorithm proposed in this paper effectively mitigates the issues associated with the former two algorithms, resulting in only a few unevenly distributed solutions. Hence, the NNC method is well-suited for addressing multi-objective problems and is capable of achieving a uniform Pareto front.

As observed in Table 4, although the voltage offset obtained by the ε-constrained method is smaller than that of the NNC method by 2.59, the distribution network cost is 4628.51RMB higher, resulting in an increased economic cost. Both the voltage offset and distribution network cost derived from the NSGA-II algorithm are larger than those of the NNC algorithm. Thus, the NSGA-II algorithm's solution is inferior to that of the NNC algorithm in terms of voltage quality and economy.

Table 4
Results of various algorithms.

Algorithm	Voltage offset /pu	Cost of distribution network /RMB
NNC method	44.60	34 806.02
ε constraint method	42.01	39 434.53
NSGA – II	43.74	39 841.57

Consequently, the NNC algorithm effectively considers the system voltage distribution and economy, ensuring safe and cost-effective system operation.

7. Conclusion

In this paper, a novel distribution network optimization model based on the NNC method, incorporating VPP and FL clusters, is proposed along with a solution method. This model offers a new perspective for optimizing distribution networks with VPP integration. Through the analysis and verification of cases, the following conclusions are derived:

(1) The NNC method proposed in this paper offers better distribution and uniformity compared to the NSGA – II and ε-constrained methods, providing decision-makers with a superior solution set for informed decision-making and ultimately leading to the optimal solution.

(2) The interconnection of VPP with the distribution network can compensate for the shortage of active power in the distribution network, consequently enhancing the power flow, reducing the system network loss, and decreasing operating expenses.

(3) The dispatch of FL clusters enables the optimization of FL, meeting customers' electricity demands while alleviating the pressure on power supply in the distribution network. Working in tandem with the DR model under time-of-use tariffs, the FL clusters achieve peak-shaving and valley-filling effects, resulting in the cost-effective operation of the system.

CRediT authorship contribution statement

Yuhao Li: Software, Methodology, Conceptualization, Writing – original draft, Writing – review & editing. **Xiaohui Yang:** Supervision, Funding acquisition, Project administration, Resources. **Fuwei Deng:** Software, Validation, Writing – review & editing. **Yeheng Deng:** Visualization, Validation. **Xiaopeng Wang:** Visualization, Formal analysis. **Zhonglian Zhang:** Investigation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The attached picture contains some data.

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Appendix A

See Figs. A.1–A.5.

Appendix B

See Table B.1.

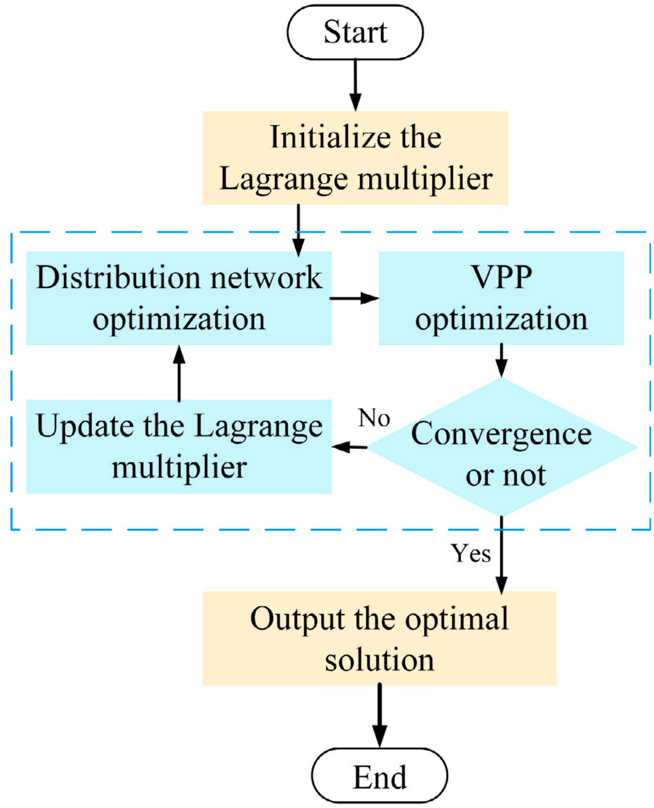

Fig. A.1. Bi-level iterative flow chart.

Table B.1
Parameter value of this paper.

Parameters	Value	Parameters	Value	Parameters	Value
E_b	30 kWh	q_{ph}	0.35 RMB /kW	η_{ec}	4
l_{max}	300 km	q_{csp}	0.03 RMB /kW	η_{re}	80%
p_{cha}	6 kW	q_{gt}	0.03 RMB /kW	η_{ex}	90%
R_a	2.268 °C/kW	q_{re}	0.02 RMB /kW	η_{ac}	1.2
γ_{AC}	3.15	q_{ex}	0.025 RMB /kW	η_{ggt}	35%
q^{PV}	0.35 RMB/kW	q_{gb}	0.02 RMB /kW	η_{gb}	90%
l^{PV}	0.35 RMB /kW	q_{eb}	0.02 RMB /kW	η_{csp}	0.43
q^{WT}	0.8 RMB /kW	q_{ec}	0.01 RMB /kW	η_{gt}	1.47
l^{WT}	0.8 RMB /kW	q_{ac}	0.025 RMB /kW	η_{eb}	0.95
U_{max}	1.05 pu	ρ	1000 kg/m^3	η_{cha}	0.9
U_{min}	0.95 pu	g	9.8 N/kg	r^{mt}_{up}	600 kW
q_{gas}	9.78 kW/ m^3	H	150 m	r^{mt}_{down}	−600 kW
W^{csp}_{min}	1.5 × 10^4 kW	ζ	0.1	W^{ph}_{min}	10 × 10^4 m^3
W^{csp}_{max}	1.5 × 10^5 kW	λ_{gas}	2.94 RMB /m^3	W^{ph}_{max}	10 × 10^5 m^3

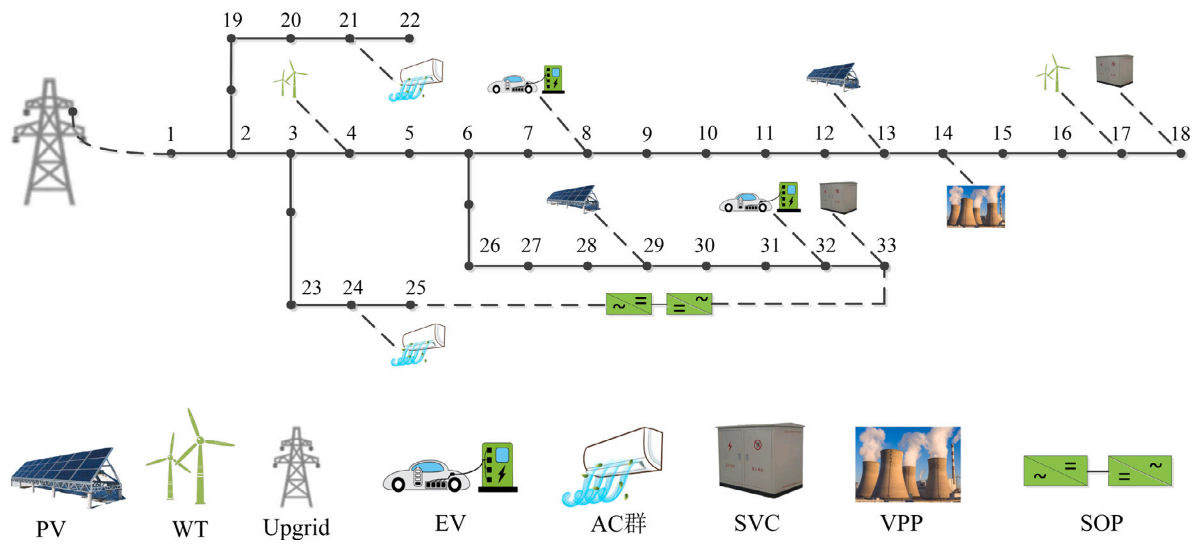

Fig. A.2. Improved IEEE33 node distribution network.

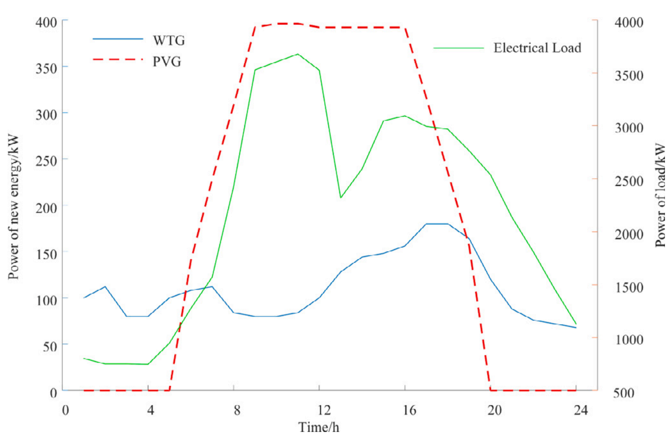

Fig. A.3. Distribution network load and new energy forecast curve.

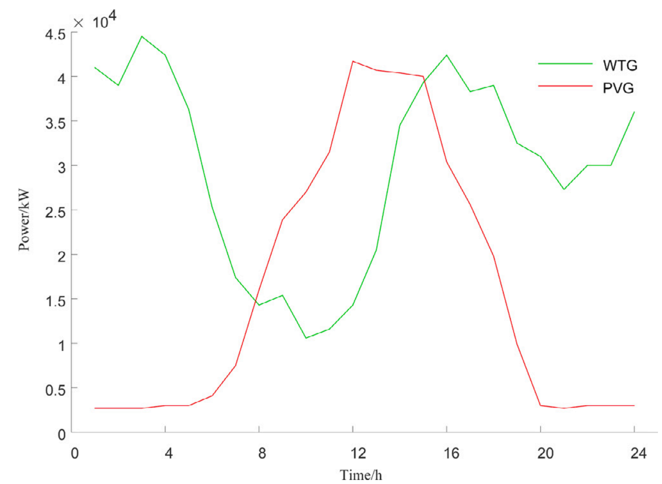

Fig. A.5. New energy prediction curve in VPP.

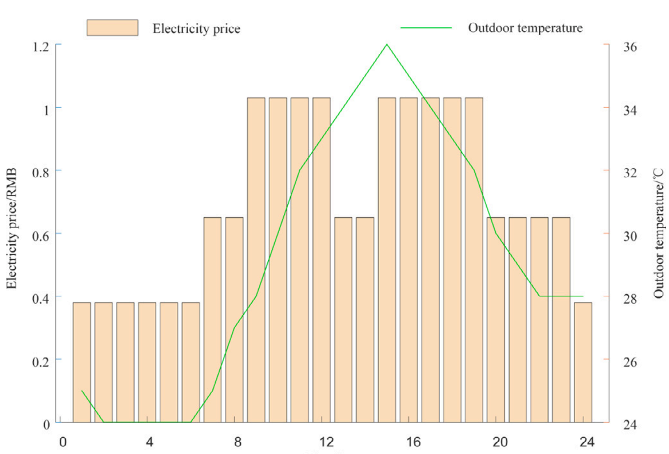

Fig. A.4. Peak and valley electricity price and outdoor temperature prediction curve.

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